ray emission observed by LAT is probably negligible.

Furthermore, the pulsar wind interaction model of [28] predicts gamma-ray spectra that extend well above 1 GeV into the TeV domain, with possible spectral turnovers and breaks above ∼100 GeV. These high cutoff energies are at odds with our observed spectral break energy in the GeV range. To explain a GeV spectral break in the pulsar wind interaction model, the maximum energy of the accelerated particles should be limited to a few GeV; this scale is below the injection energies expected for electrons and positrons from the inner pulsar magnetospheres, which may range up to TeV energies [29]. Consequently, pulsar wind interactions should play a minor role in the acceleration of electrons and positrons in 47 Tuc. This is consistent with the model of [24], which suggests that the direct conversion efficiency of spin-down energy into gamma rays, η_γ, is considerably larger than the efficiency $\eta_{e\pm}$ for electron and positron production. We cannot exclude, however, the possibility that pulsar wind interactions contribute at a low level to the gamma-ray signal we detected from 47 Tuc. Because TeV gamma-ray emission from 47 Tuc has not yet been detected [30], we cannot place firm constraints on that contribution.

Until now the study of close binary systems in globular clusters has mainly relied on X-ray observations. Such studies, however, are hampered by the fact that a large variety of binary systems emit X-rays [cataclysmic variables (CVs), low-mass x-ray binaries (LMXBs), chromospherically active main-sequence binaries (BY Dras/RS CVns), and MSPs] and that it is difficult to assess the nature of the sources from X-ray observations alone (however, see [11]). X-ray studies must therefore be backed up by multiwavelength identification programs that help to disentangle these source populations. High-energy gamma-ray observations are unique in that they should be sensitive mainly to the pulsar populations. This is illustrated by Fermi observations of our own Galaxy that have revealed that pulsars form the largest and most luminous point-source population in this energy domain. No CVs, LMXBs, or BY Dras have so far been detected in high-energy gamma rays [31]. It thus seems rather likely that pulsars (and MSPs in particular) are also the primary population of gamma-ray sources in globular clusters.

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